Saransha Basu

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Machine Learning & AI Student | Teaching Assistant Applicant

Computer Engineering student focused on machine learning, deep learning, NLP, and practical AI systems. Builds models and learning tools from first principles, with hands-on work in NumPy, PyTorch, Python, semantic search, sequence modelling, and audio classification. Enjoys mentoring classmates and explaining ML/DL concepts through structured notes and intuitive derivations.

EDUCATION

Bachelor in Computer Engineering — Khwopa Engineering College, Bhaktapur, Nepal Expected 2027

Cumulative GPA: 3.62 / 4.00 after four semesters (equal-credit calculation)

Semester GPAs: 3.56, 3.58, 3.64, 3.71 · Completed credits: 21, 18, 17, 18

SELECTED PROJECTS

Autograd Engine from Scratch GitHub

- Implemented a pure-NumPy reverse-mode automatic differentiation engine without PyTorch, TensorFlow, or autograd libraries.
- Built dynamic computation graphs, gradient accumulation, reverse traversal, matrix multiplication, reductions, broadcasting, and a two-layer network trained on XOR.

Technologies: Python, NumPy, automatic differentiation, computational graphs, SGD

BIRDCLEF 2025: CNN, Attention, and AST Comparison GitHub

- Explored bird-audio classification using three approaches: a conventional CNN, a hybrid CNN with attention, and an Audio Spectrogram Transformer with attention.
- Compared model behaviour under different data regimes, observing that CNNs can be effective with limited data while AST models benefit from larger datasets.

Technologies: Python, Jupyter, deep learning, CNNs, attention, ASTs, audio classification

Semantic-Note GitHub

- Completed a FastAPI application for storing notes and retrieving semantically related content using sentence embeddings.
- Organized the application into API routes, database models, embedding services, search logic, migrations, tests, and a Streamlit interface; not yet deployed.

Technologies: Python, FastAPI, sentence embeddings, SQLAlchemy, semantic search, Streamlit

50 ML Projects to Understand LLMs GitHub

- Built a structured collection of focused implementations to study the foundations, architectures, and techniques behind modern language models.
- Used the repository as a practical learning portfolio connecting theory, experimentation, and reproducible notebooks.

Technologies: Python, machine learning, deep learning, NLP, Jupyter

Additional ML and NLP Projects Portfolio

Character-level RNN question-answering system; next-word predictor; implementations of linear/logistic regression, decision trees, K-means, PCA, gradient descent, and regularization from scratch.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Data: NumPy, Pandas, Matplotlib, Seaborn

Databases: PostgreSQL, MySQL, SQLite, MongoDB

ML/DL: PyTorch, TensorFlow, Keras, scikit-learn

Web/API: FastAPI, Flask, React, Next.js

Tools: Git, Docker, Linux, Jupyter

TEACHING & LEADERSHIP

- Mentored classmates in ML fundamentals, deep-learning fundamentals, Python, NumPy, Pandas, and NLP.
- Prepared 15+ sets of personal learning notes covering machine learning and deep learning from first principles.
- Interested in supporting hands-on labs by explaining theory, implementation details, debugging approaches, and practical model-building decisions.

PROFILE

Engineering student · AI/ML learner · Interested in deep learning, NLP, computer vision, and building practical AI systems.